

Midterm Exam Discussion

- ▶ I know the midterm exam questions are indeed too difficult for your current level.
- ▶ There are two reasons for this:
 - ▶ 1. The scope of the questions is too broad and too difficult.
 - ▶ 2. Due to the closed-book exam.

Midterm Exam Discussion

- ▶ Therefore, I will first improve everyone's grades and uniformly add points for all students.

- ▶ Example:

Final midterm exam score

= original midterm exam score + 20 points

- ▶ Therefore, please refer to the current revised grades.

Midterm Exam Discussion

- ▶ Afterwards, the final exam will be replaced with a **final reflection report**.
- ▶ Therefore, please submit your assignments on time.
- ▶ There are still **5** homework assignments remaining.
- ▶ We hope everyone will cooperate, OK?
- ▶ The final course grade =
Midterm Exam (30%) + HW (30%) + Final Report (30%) + Attendance (10%)

Final Report Topic

- ▶ 1. Topic: I will provide 16 topics for you to choose from.
- ▶ 2. Font Size: 12.
- ▶ 3. Submission Date: Monday of the last week of the semester. Email me.
- ▶ 4. Number of Pages: 8-10

16 topics list (1)

- ▶ 1. The Evolution of CPU Architecture: Development and Challenges from Single-Core to Multi-Core Processors
- ▶ 2. Cache Memory Design: Analysis of the Impact of Hierarchical Structure on System Performance
- ▶ 3. Application of Pipeline Technology in Modern Processors: Advantages, Risks, and Solutions
- ▶ 4. Comparison of RISC and CISC Architectures: Performance, Power Consumption, and Application Scenarios

16 topics list (2)

- ▶ 5. GPU Hardware Architecture: Parallel Computing Principles and Their Role in Graphics Processing
- ▶ 6. Memory Hierarchy Optimization: Design Strategies from Scratchpads to Virtual Memory
- ▶ 7. Quantum Computing Hardware Architecture: Implementation Challenges of Quantum Bits and Gates
- ▶ 8. The End of Moore's Law: Trends and Innovations in Future Computer Hardware Architecture

16 topics list (3)

- ▶ 9. Comparison and Future Development of RISC vs. CISC Processor Architectures: This section explores the design principles, performance differences, and advantages and disadvantages of the two architectures in modern applications.
- ▶ 10. Design and Optimization of Cache Hierarchy: This section analyzes the hierarchical structure of cache memory, hit rate calculation, and how hardware design can improve system performance.
- ▶ 11. Applications of Multi-Core Processors in Parallel Computing: This section discusses the architecture of multi-core CPUs, shared resource management, and the challenges and solutions in high-performance computing.
- ▶ 12. GPU Architecture and Graphics Processing Applications: This section introduces the parallel processing architecture of GPUs, hardware support for CUDA or OpenCL, and their roles in AI and games.

16 topics list (4)

- ▶ 13. Embedded System Hardware Architecture Design: This section focuses on ARM-based embedded processors, power management, and practical examples in IoT devices.
- ▶ 14. Hardware Implementation of Virtual Memory and Page Replacement Algorithms: This section explores the hardware support for virtual memory, the role of TLBs, and the performance impact of different replacement strategies.
- ▶ 15. Fundamentals and Challenges of Quantum Computing Hardware Architecture: Introduces the hardware implementation of qubits and quantum gates, as well as the current architectural limitations of quantum computers.
- ▶ 16. Architecture Analysis of AI-Dedicated Hardware Accelerators (such as TPUs): Discusses the design principles of tensor processing units, their comparison with traditional CPUs/GPUs, and their advantages in deep learning.